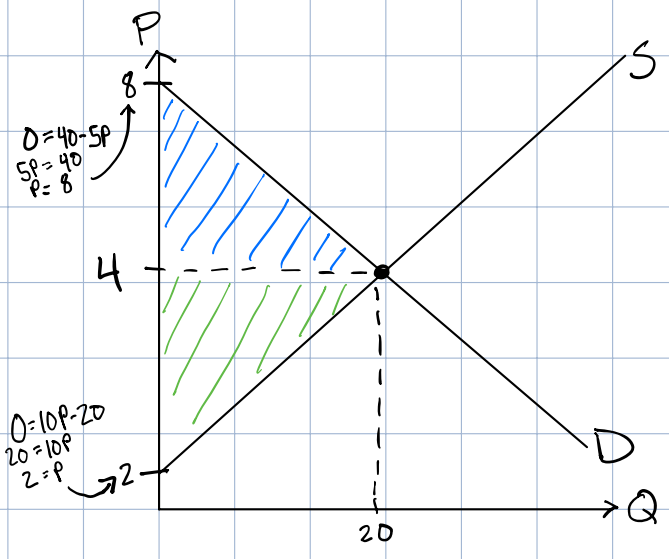
Solns

1. a.



$40-5P^* = 10P^* - 20$

$60 = 15P^*$

$4 = P^*$

$Q^* = 40-5(4)$

$= 20$

b. $CS = \frac{1}{2}(8-4)(20) = 40$

$PS = \frac{1}{2}(4-2)(20) = 20$

2. a. $105-15P^* = 25P^* - 15$

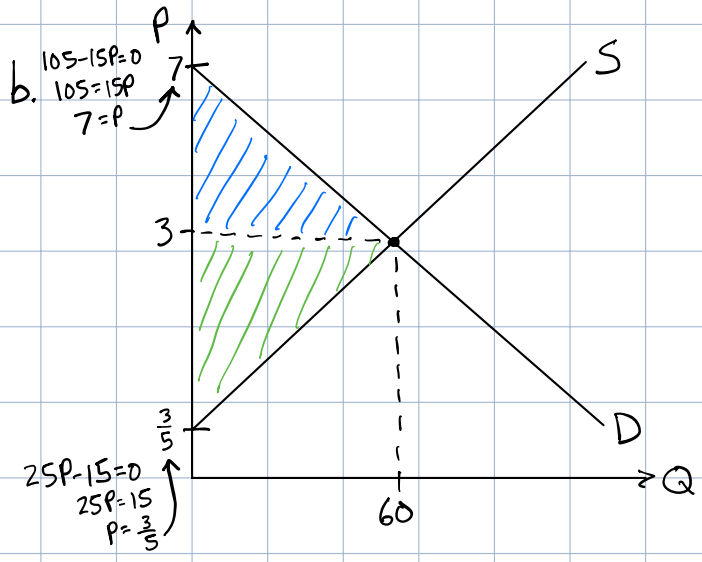
$120 = 40P^*$

$3 = P^*$

$Q^* = 105-15(3)$

$= 60$

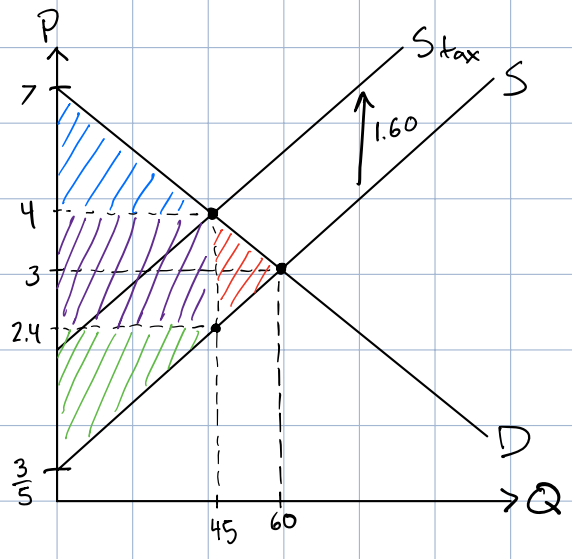
b.



$CS = \frac{1}{2}(7-3)(60) = 120$

$PS = \frac{1}{2}(3-\frac{3}{5})(60) = \frac{1}{2}(\frac{12}{5})(60) = 72$

c.



$105-15P_{consumer} = 25(P_{consumer}-1.60)-15$

$105-15P_c = 25P_c - 40 - 15$

$160 = 40P_c$

$4 = P_c$

$P_{supplier} = 4-1.6$

$= 2.4$

$Q_{tax} = 105-15(4)$

$= 45$

d. $P_{consumer} = 4$ $P_{producer} = 2.4$ (See work above)

e. $CS = \frac{1}{2}(7-4)(45) = \frac{135}{2}$

$PS = \frac{1}{2}(2.4-0.6)(45) = \frac{1}{2}\left(\frac{12}{5}-\frac{3}{5}\right)(45) = \frac{9}{5}\cdot\frac{45}{2} = \frac{81}{2}$

f. $Tax\ Rev = 1.6(45) = 72$

g. $DWL = \frac{1}{2}(4-2.4)(15) = 0.8(15) = 12$